

Team : Visionary, AI

Q11) Answer $\Rightarrow$ A) John has a teaching license.

Explanation :

Given Claim 1 $\#$: $(\forall x (\text{Teacher}(x) \rightarrow \text{HasLicense}(x)))$

Claim 2 : $\# \text{Teacher}(\text{John})$

Let $x = \text{John}$, substitute to Claim 1 :

$\# \text{Teacher}(\text{John}) \rightarrow \text{HasLicense}(\text{John})$

Use Modus Ponens rule (where $P \rightarrow Q$, then Q)

$\text{HasLicense}(\text{John})$

$\therefore$ Thus, John has license.

Q12) Answer $\Rightarrow$ () A valid conclusion that accurately reflects original premises.

Explanation :

- $\hookrightarrow$ ~~B~~ Because we start from assume "False", and show it is wrong, thus it prove that "False" not exist,
- $\hookrightarrow$ A is wrong as we use complete ~~in~~ data throughout the process, just we assume "False" statement.
- $\hookrightarrow$ B is tricky wrong, as NIL resolve the "True" statement as it show "False" didn't exist.
- $\hookrightarrow$ D wrong as NIL is highly relevant to prove "True" exist.

Q13 Alice is fit

Fact

- 1 Person (Alice)
- 2 Exercise (Alice)
- 3 Person (Bob)

Rule $\rightarrow$ Person(x)
 $\searrow$ Exercises(x)

Rule 1 Person (Alice)
X=Alice Exercises(Alice)
match Rule 1

Rule 1
x=Bob dont have Exercise(Bob)

AND CONDITION not met Rule 1 not for Bob

Rule 2

x=Alice Person(Alice). Rule fire
y=Bob Person(Bob). the rule fire

Q14 LIFO

$\uparrow$
DFS

1) call PF5(node) $\xrightarrow{\text{call}}$ DFS(neighbor) $\rightarrow$ on top of call stack
 $\downarrow$
deep call

when no visited (dead end) it finish
it pop up

No.:

Question 15:

Answer: C) I, II, IV

A blind search Algorithm is one that explores
state space
a problem space ~~state~~ without any extra information.

I - Uniform cost search is a blind search

III. Breadth-First search is a blind search.

IV. Depth-first search is a blind search.

Hence: I, II, IV which is answer C.

Question 16:

Answer: B) Breadth-First search

Because it uses a Queue data structure, which
operates on a first-in, first out principle.



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Q17- (c) Uniform cost search

GPS/Route Finding operates on weighted graphs. which is why uniform cost search is the best algorithm for finding nearby locations and calculating the ~~cheapest~~ ~~least~~ least costing route.

Q18- (d) Heuristic search

Algorithms that use knowledge of the destination (A^*) are heuristic which estimate the distance to decide which direction to go next.

Like how far you have gone (g) and how far you still need to go (h)

Then it picks the best route $F = g + h$